STUDENT TEAM WEEKLY REPORT: MODULE 5

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [50+ Client Ongoing Test Handling, Login Attempt TIMEOUT & Author Concurrency Lock]	**TEAM ID:** 3 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 07/09/2026 **STUDENT LEADER:** Pranav Chougule

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• **Login Attempt Rate Limiting & TIMEOUT:** Built LoginRateLimiter.java enforcing a 5-failed-attempt threshold within a 15-minute sliding window; returns HTTP 429 Too Many Requests with remaining lockout countdown. • **Ongoing Test Handling for 50+ Clients:** Scaled AppServer.java with thread pooling and non-blocking I/O; tested with simulate_100_quiz_students.py executing 100 simultaneous active quiz submissions without drops. • **Concurrent Submission of QBs by Authors:** Implemented SubjectConcurrencyManager.java using ConcurrentHashMap<String, ReentrantLock> to enforce thread-safe question bank uploads for the same subject. • **Concurrent Author Benchmark:** Built simulate_concurrent_authors.py simulating 5 faculty members submitting question banks for the same subject (Data Science) at the exact same millisecond. • **Lockout Demo Reset Utility:** Implemented POST /api/auth/reset-rate-limit for evaluation and testing.	• **Subject Quota Race Conditions:** Simultaneous faculty uploads for the same course caused overwritten batches and quota conflicts. Action: Wrapped question bank saving inside fair ReentrantLock.executeWithLock(). **GREEN (RESOLVED)** • **Brute-Force & Denial-of-Service:** Repeated password guessing threatened server stability. Action: Enforced 5-attempt threshold with automatic 15-minute lockout timer. **GREEN (RESOLVED)** • **High Latency Under 50+ Concurrent Clients:** Thread pool contention caused response delays during mass test submission. Action: Optimized thread pool sizing to Executors.newCachedThreadPool(). **GREEN (RESOLVED)**	**Distributed Network Testing:** Request the Integration team to execute simulate_100_quiz_students.py across multiple client machines simultaneously to benchmark real Wi-Fi network throughput.

INTEGRATION AND COLLABORATION
• **Integration Team:** Validated multi-threaded concurrency scripts against AppServer.java on port 8080. • **Frontend Team:** Integrated HTTP 429 JSON response payload (remainingSeconds, locked: true) with #lockoutModal on login.html. • **Database Team:** Tuned PostgreSQL connection pool max connections to support high-concurrency student quiz submissions. • **Pipeline Learnings:** Mastered multi-threaded thread synchronization, reentrant locks, and high-concurrency performance engineering.

COMMON KPI EVIDENCES
50+ Student Test Concurrency: Handled 100 simultaneous quiz submissions (double the 50-client requirement) with 100% success (100/100), 1,089.9 RPS, and average latency 79.67 ms.
Concurrent Author Uploads: 5 simultaneous faculty uploads executed with 0 lock errors, 0 race conditions, and total execution time 0.01s.
Rate-Limiting Protection: HTTP 429 status and 15-minute countdown accurately triggered on the 5th failed attempt.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• **Pranav Chougule:** Develop O(1) auto-grading engine for objective MCQ and True/False questions. (Due: Sep 12, 2026) • **Pranav Chougule:** Build backend manual evaluation endpoint (/api/quiz-results/grade) for subjective answers. (Due: Sep 17, 2026)	